Assignment 01

Starting Client Side:

```
C:\Users\PC>mysql -u d6_96728_shaurya -p --system-command
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 25
Server version: 9.6.0 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

```
d6_96728_shaurya> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| classwork          |
| information_schema |
| mysql              |
| performance_schema |
| spj                |
| sys                |
| vru                |
+--------------------+
7 rows in set (0.012 sec)

d6_96728_shaurya>
```

DATABASE Changed:

```
d6_96728_shaurya>SHOW TABLES;
+---------------+
| Tables_in_spj |
+---------------+
| p             |
| s             |
+---------------+
2 rows in set (0.026 sec)

d6_96728_shaurya>
```

```
d6_96728_shaurya>DESCRIBE p;
+--------+----------+------+-----+---------+-------+
| Field  | Type     | Null | Key | Default | Extra |
+--------+----------+------+-----+---------+-------+
| P#     | char(6)  | NO   |     | NULL    |       |
| Pname  | char(20) | YES  |     | NULL    |       |
| color  | char(6)  | YES  |     | NULL    |       |
| weight | smallint | YES  |     | NULL    |       |
| city   | char(15) | YES  |     | NULL    |       |
+--------+----------+------+-----+---------+-------+
5 rows in set (0.020 sec)

d6_96728_shaurya>DESCRIBE s;
+--------+----------+------+-----+---------+-------+
| Field  | Type     | Null | Key | Default | Extra |
+--------+----------+------+-----+---------+-------+
| S#     | char(5)  | NO   |     | NULL    |       |
| Sname  | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
| city   | char(15) | YES  |     | NULL    |       |
+--------+----------+------+-----+---------+-------+
4 rows in set (0.014 sec)

d6_96728_shaurya>
```

Creating and Populating J:

```
d6_96728_shaurya>INSERT INTO J (`J#`, Jname, City) VALUES
    -> ('J1', 'Sorter', 'Paris'),
    -> ('J2', 'Punch', 'Rome'),
    -> ('J3', 'Reader', 'Athens'),
    -> ('J4', 'Console', 'Athens'),
    -> ('J5', 'Collator', 'London'),
    -> ('J6', 'Terminal', 'Oslo'),
    -> ('J7', 'Tape', 'London');
Query OK, 7 rows affected (0.061 sec)
Records: 7  Duplicates: 0  Warnings: 0

d6_96728_shaurya>
```

```
d6_96728_shaurya>INSERT INTO sp (`S#`, `P#`, `J#`, QTY) VALUES
    -> ('S1', 'P1', 'J1', 200),
    -> ('S1', 'P1', 'J4', 700),
    -> ('S2', 'P3', 'J1', 400),
    -> ('S2', 'P3', 'J2', 200),
    -> ('S2', 'P3', 'J3', 200),
    -> ('S2', 'P3', 'J4', 500),
    -> ('S2', 'P3', 'J5', 600),
    -> ('S2', 'P3', 'J6', 400),
    -> ('S2', 'P3', 'J7', 800),
    -> ('S2', 'P5', 'J2', 100),
    -> ('S3', 'P3', 'J1', 200),
    -> ('S3', 'P4', 'J2', 500),
    -> ('S4', 'P6', 'J3', 300),
    -> ('S4', 'P6', 'J7', 300),
    -> ('S5', 'P2', 'J2', 200),
    -> ('S5', 'P2', 'J4', 100),
    -> ('S5', 'P5', 'J5', 500),
    -> ('S5', 'P5', 'J7', 100),
    -> ('S5', 'P6', 'J2', 200),
    -> ('S5', 'P1', 'J4', 100),
    -> ('S5', 'P3', 'J4', 200),
    -> ('S5', 'P4', 'J4', 800),
    -> ('S5', 'P5', 'J4', 400),
    -> ('S5', 'P6', 'J4', 500);
Query OK, 24 rows affected (0.345 sec)
Records: 24  Duplicates: 0  Warnings: 0

d6_96728_shaurya>
```

Importing HR database:

```
d6_96728_shaurya>SOURCE C:/Users/PC/Desktop/CDAC_Sunbeam/Class/DataBase/dbs/new-hr-db.sql;
Query OK, 0 rows affected (0.003 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.001 sec)

Query OK, 0 rows affected (0.001 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.001 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.002 sec)
```

Importing Sales database:

```
d6_96728_shaurya>USE sales;
Database changed
d6_96728_shaurya>SOURCE C:/Users/PC/Desktop/CDAC_Sunbeam/Class/DataBase/dbs/sales-db.sql;
Query OK, 0 rows affected, 1 warning (0.227 sec)

Query OK, 0 rows affected, 3 warnings (0.193 sec)

Query OK, 0 rows affected, 3 warnings (0.195 sec)

Query OK, 1 row affected (0.034 sec)

Query OK, 1 row affected (0.040 sec)

Query OK, 1 row affected (0.038 sec)

Query OK, 1 row affected (0.047 sec)

Query OK, 1 row affected (0.038 sec)

Query OK, 1 row affected (0.064 sec)
```

Viewing HR Tables

```
d6_96728_shaurya>USE hr;
Database changed
d6_96728_shaurya>SHOW TABLES;
+---------------+
| Tables_in_hr |
+---------------+
| countries     |
| departments   |
| employees     |
| job_history   |
| jobs          |
| locations     |
| regions       |
+---------------+
7 rows in set (0.009 sec)
```

Table: Employee

```
d6_96728_shaurya>SELECT * FROM employees;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60
104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60
105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	4800.00	NULL	103	60
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_PROG	4200.00	NULL	103	60
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR	12008.00	NULL	101	100
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
110	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_ACCOUNT	8200.00	NULL	108	100
111	Ismael	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	7700.00	NULL	108	100
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_ACCOUNT	7800.00	NULL	108	100
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	6900.00	NULL	108	100
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30
116	Shelli	Baida	SBAIDA	1.515.555.0116	2015-12-24	PU_CLERK	2900.00	NULL	114	30
117	Sigal	Tobias	STOBIAS	1.515.555.0117	2015-07-24	PU_CLERK	2800.00	NULL	114	30
118	Guy	Himuro	GHIMURO	1.515.555.0118	2016-11-15	PU_CLERK	2600.00	NULL	114	30

Viewing HR Tables:

```
d6_96728_shaurya>USE hr;
Database changed
d6_96728_shaurya>SELECT * FROM countries;
+------------+----------------------------------------+-----------+
| COUNTRY_ID | COUNTRY_NAME                           | REGION_ID |
+------------+----------------------------------------+-----------+
| AR         | Argentina                              |        20 |
| AU         | Australia                              |        40 |
| BE         | Belgium                                |        10 |
| BR         | Brazil                                 |        20 |
| CA         | Canada                                 |        20 |
| CH         | Switzerland                            |        10 |
| CN         | China                                  |        30 |
| DE         | Germany                                |        10 |
```

TABLE: departments

```
d6_96728_shaurya>SELECT * FROM departments;
+---------------+----------------------+------------+-------------+
| DEPARTMENT_ID | DEPARTMENT_NAME      | MANAGER_ID | LOCATION_ID |
+---------------+----------------------+------------+-------------+
|            10 | Administration       |        200 |        1700 |
|            20 | Marketing            |        201 |        1800 |
|            30 | Purchasing           |        114 |        1700 |
|            40 | Human Resources      |        203 |        2400 |
|            50 | Shipping             |        121 |        1500 |
```

TABLE : employees

```
d6_96728_shaurya>SELECT * FROM employees;
+-------------+------------+-----------+----------+--------------+------------+----------+---------+----------------+------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL    | PHONE_NUMBER | HIRE_DATE  | JOB_ID   | SALARY  | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+----------+--------------+------------+----------+---------+----------------+------------+---------------+
|         100 | Steven     | King      | SKING    | 1.515.555.0100 | 2013-06-17 | AD_PRES  | 24000.00 |           NULL |       NULL |            90 |
|         101 | Neena      | Yang      | NYANG    | 1.515.555.0101 | 2015-09-21 | AD_VP    | 17000.00 |           NULL |        100 |            90 |
|         102 | Lex        | Garcia    | LGARCIA  | 1.515.555.0102 | 2011-01-13 | AD_VP    | 17000.00 |           NULL |        100 |            90 |
|         103 | Alexander  | James     | AJAMES   | 1.590.555.0103 | 2016-01-03 | IT_PROG  |  9000.00 |           NULL |        102 |            60 |
|         104 | Bruce      | Miller    | BMILLER  | 1.590.555.0104 | 2017-05-21 | IT_PROG  |  6000.00 |           NULL |        103 |            60 |
|         105 | David      | Williams  | DWILLIAMS | 1.590.555.0105 | 2015-06-25 | IT_PROG  |  4800.00 |           NULL |        103 |            60 |
```

TABLE: job_history

```
d6_96728_shaurya>SELECT * FROM job_history;
+-------------+------------+------------+------------+---------------+
| EMPLOYEE_ID | START_DATE | END_DATE   | JOB_ID     | DEPARTMENT_ID |
+-------------+------------+------------+------------+---------------+
|         101 | 2007-09-21 | 2011-10-27 | AC_ACCOUNT |           110 |
|         101 | 2011-10-28 | 2015-03-15 | AC_MGR     |           110 |
|         102 | 2011-01-13 | 2016-07-24 | IT_PROG    |            60 |
|         114 | 2016-03-24 | 2017-12-31 | ST_CLERK   |            50 |
```

TABLE: jobs

```
d6_96728_shaurya>SELECT * FROM jobs;
+------------+----------------------------------+------------+------------+
| JOB_ID     | JOB_TITLE                        | MIN_SALARY | MAX_SALARY |
+------------+----------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant                |       4200 |       9000 |
| AC_MGR     | Accounting Manager               |       8200 |      16000 |
| AD_ASST    | Administration Assistant         |       3000 |       6000 |
| AD_PRES    | President                        |      20080 |      40000 |
```

TABLE: locations

```
d6_96728_shaurya>SELECT * FROM locations;
+------------+---------------------------------------+------------+--------------------+-----------------+------------+
| LOCATION_ID | STREET_ADDRESS                        | POSTAL_CODE | CITY               | STATE_PROVINCE  | COUNTRY_ID |
+------------+---------------------------------------+------------+--------------------+-----------------+------------+
|        1000 | 1297 Via Cola di Rie                  | 00989      | Roma               | NULL            | IT         |
|        1100 | 93091 Calle della Testa               | 10934      | Venice             | NULL            | IT         |
|        1200 | 2017 Shinjuku-ku                      | 1689       | Tokyo              | Tokyo Prefecture | JP         |
|        1300 | 9450 Kamiya-cho                       | 6823       | Hiroshima          | NULL            | JP         |
```

TABLE: regions

```
d6_96728_shaurya>SELECT * FROM regions;
+-----------+-------------+
| REGION_ID | REGION_NAME |
+-----------+-------------+
|        50 | Africa      |
|        20 | Americas    |
|        30 | Asia        |
|        10 | Europe      |
|        40 | Oceania     |
+-----------+-------------+
5 rows in set (0.003 sec)
```

Viewing sales Tables:

TABLE: customers

```
d6_96728_shaurya>SELECT * FROM customers;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2001 | Hoffman  | London   |    100 | 1001 |
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2006 | Clemens  | London   |    100 | 1001 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
| 2007 | Pereira  | Rome     |    100 | 1004 |
+------+----------+----------+--------+------+
7 rows in set (0.003 sec)
```

TABLE: orders

```
d6_96728_shaurya>SELECT * FROM orders;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.003 sec)
```

TABLE: salespeople

```
d6_96728_shaurya>SELECT * FROM salespeople;
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.12 |
| 1002 | Serres  | San Jose  | 0.13 |
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
5 rows in set (0.003 sec)
```

Observations:

1. How many orders data is present in the database?
   There are 10 orders present in the orders table.

2. How many customers are present in the database?
   There are 7 customers present in the customers table.

3. How many salespeople are present in the database?
   There are 5 salespeople present in the salespeople table.

4. On which date order with highest amount is placed?
   The highest amount order ($9891.88) was placed on 1990-10-04.

5. For which order salesman got maximum commission?
   The maximum commission was earned for Order 3011 (Commission: 1187.0256).

6. Which salesmen are handling more than two customers?
   None. According to the data, no single salesman is currently handling more than two customers (all handle 1 or 2).

7. Which city have single customer?
   Berlin is the only city with exactly one customer.

8. Which city has multiple salespeople?
   London has multiple salespeople.

9. What is name of customer and salesman of the maximum amount order?
   The customer is Clemens and the salesman is Peel for the order amounting to $9891.88.

10. Study relationship between the tables in sales database.

    Primary Keys: snum (Salespeople), cnum (Customers),
    and onum (Orders) uniquely identify rows in their respective tables.

    Foreign Keys: The orders table is the "child" to
    both customers and salespeople, meaning it depends on them existing first.